

# ArtGAN: WikiArt Model Training

Follow these steps to train your CNN-RNN hybrid model on the 25GB WikiArt dataset using Colab's free GPU.

```
In [ ]: # After mounting drive
!cp /content/drive/MyDrive/ArtGAN_Code/*.py /content/
!cp /content/drive/MyDrive/ArtGAN_Code/'WikiArt Dataset.zip' /content/
!unzip -q 'WikiArt Dataset.zip' -d /content/
```

```
In [ ]: from google.colab import drive
drive.mount('/content/drive')
```

Drive already mounted at /content/drive; to attempt to forcibly remount, call `drive.mount("/content/drive", force_remount=True)`.

```
In [ ]: # 1. Mount Drive & Setup Folders
from google.colab import drive
import os
drive.mount('/content/drive')

# Create persistent folder for your model results
DRIVE_BASE = '/content/drive/MyDrive/ArtGAN_Results'
os.makedirs(DRIVE_BASE, exist_ok=True)

# 2. Install dependencies
!pip install tqdm pandas pillow torch torchvision
```

Mounted at /content/drive  
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (4.67.3)  
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)  
Requirement already satisfied: pillow in /usr/local/lib/python3.12/dist-packages (11.3.0)  
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cu128)  
Requirement already satisfied: torchvision in /usr/local/lib/python3.12/dist-packages (0.25.0+cu128)  
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)  
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)  
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)  
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)  
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)  
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)  
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)  
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)  
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)  
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)  
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)  
Requirement already satisfied: cuda-bindings==12.9.4 in /usr/local/lib/python3.12/dist-packages (from torch) (12.9.4)  
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.93)  
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)  
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.90)  
Requirement already satisfied: nvidia-cudnn-cu12==9.10.2.21 in /usr/local/lib/python3.12/dist-packages (from torch) (9.10.2.21)  
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.4.1)  
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83 in /usr/local/lib/python3.12/dist-packages (from torch) (11.3.3.83)  
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90 in /usr/local/lib/python3.12/dist-packages (from torch) (10.3.9.90)  
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90 in /usr/local/lib/python3.12/dist-packages (from torch) (11.7.3.90)  
Requirement already satisfied: nvidia-cusparselt-cu12==12.5.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12.5.8.93)  
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12.5.8.93)  
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/lib/python3.12/dist-packages (from torch) (0.7.1)  
Requirement already satisfied: nvidia-nccl-cu12==2.27.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2.27.5)  
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/python3.12/dist-packages (from torch) (3.4.5)  
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90 in /usr/local/lib/python

3.12/dist-packages (from torch) (12.8.90)  
 Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93 in /usr/local/lib/python3.12/dist-packages (from torch) (12.8.93)  
 Requirement already satisfied: nvidia-cublas-cu12==11.11.3.6 in /usr/local/lib/python3.12/dist-packages (from torch) (11.11.3.6)  
 Requirement already satisfied: nvidia-cuda-runtime-cu12==12.0.102 in /usr/local/lib/python3.12/dist-packages (from torch) (12.0.102)  
 Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.0.102 in /usr/local/lib/python3.12/dist-packages (from torch) (12.0.102)  
 Requirement already satisfied: nvidia-cudnn-cu12==8.9.7.29 in /usr/local/lib/python3.12/dist-packages (from torch) (8.9.7.29)  
 Requirement already satisfied: nvidia-cufft-cu12==11.0.2.53 in /usr/local/lib/python3.12/dist-packages (from torch) (11.0.2.53)  
 Requirement already satisfied: nvidia-curand-cu12==10.3.2.106 in /usr/local/lib/python3.12/dist-packages (from torch) (10.3.2.106)  
 Requirement already satisfied: nvidia-cusolver-cu12==11.6.1.2 in /usr/local/lib/python3.12/dist-packages (from torch) (11.6.1.2)  
 Requirement already satisfied: nvidia-cusparse-cu12==12.1.0.106 in /usr/local/lib/python3.12/dist-packages (from torch) (12.1.0.106)  
 Requirement already satisfied: nvidia-nccl-cu12==2.20.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2.20.5)  
 Requirement already satisfied: nvidia-nvtx-cu12==12.0.102 in /usr/local/lib/python3.12/dist-packages (from torch) (12.0.102)  
 Requirement already satisfied: sympy==1.12.0 in /usr/local/lib/python3.12/dist-packages (from torch) (1.12.0)  
 Requirement already satisfied: mpmath<1.4, >=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy==1.12.0->torch) (1.3.0)  
 Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)

In [ ]: `%cd /content/wikiart_data`

```
# Copy from your Drive to Local Colab storage (much faster for training)
!cp /content/drive/MyDrive/wikiart.zip .

# Extract
!unzip -q wikiart.zip -d .

# Verify
!ls wikiart/ | head -5
print("Done! Images extracted.")
```

[Errno 2] No such file or directory: '/content/wikiart\_data'  
 /content  
 ./wikiart/Baroque/rembrandt\_woman-standing-with-raised-hands.jpg bad CRC 3d470143 (should be 90cccc50)  
 error: invalid compressed data to inflate ./wikiart/Post\_Impressionism/vincent-van-gogh\_1-arlesienne-portrait-of-madame-ginoux-1890.jpg  
 Abstract\_Expressionism  
 Action\_painting  
 Analytical\_Cubism  
 Art\_Nouveau\_Modern  
 Baroque  
 Done! Images extracted.

## 4. Upload your Files

On the left sidebar, click the **Folder icon** . Inside `/content/` (root folder), upload these files from your computer:

1. `train.py`
2. `model.py`
3. `data_loader.py`
4. **The WikiArt Dataset folder** (You should zip this folder on your PC first, upload `WikiArt Dataset.zip`, and run the cell below).

In [ ]: *#to check whether the wikiart is extracted or not*  
`import os`  
`path = '/content/wikiart'`  
`folders = os.listdir(path)`

```
print(f"Found {len(folders)} artist folders in wikiart/")  
print("First 5:", folders[:5])
```

Found 27 artist folders in wikiart/

First 5: ['Analytical\_Cubism', 'Post\_Impressionism', 'Color\_Field\_Painting', 'Impressionism', 'Expressionism']

```
In [ ]: # 5. Start Training  
# This will save the best model to your Google Drive automatically  
%cd /content  
!python train.py
```

```

/content
Using device: cuda
Checkpoints will be saved to: /content/drive/MyDrive/ArtGAN_Results/checkpoints
Loading data...
Loaded 11276 train images with all 3 labels.
Loaded 4707 val images with all 3 labels.
Initializing model...
Downloading: "https://download.pytorch.org/models/resnet50-11ad3fa6.pth" to /root/.cache/torch/hub/checkpoints/resnet50-11ad3fa6.pth
100% 97.8M/97.8M [00:00<00:00, 188MB/s]
Trainable params: 18,125,629 / 41,633,661 (43.5%)
/content/train.py:151: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated. Please use `torch.amp.GradScaler('cuda', args...)` instead.
    scaler = GradScaler()

```

```

Epoch [1/30]
Learning Rates -> Backbone: 1.00e-05 | Head: 1.00e-03
Training: 0% 0/353 [00:00<?, ?it/s]/content/train.py:26: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
    with autocast():
Training: 22% 76/353 [00:59<04:16, 1.08it/s]Error loading ./wikiart/Post_Impressionism/vincent-van-gogh_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file
Training: 100% 353/353 [04:22<00:00, 1.35it/s]
Validating: 0% 0/148 [00:00<?, ?it/s]/content/train.py:63: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Please use `torch.amp.autocast('cuda', args...)` instead.
    with autocast():
Validating: 100% 148/148 [01:20<00:00, 1.85it/s]
Train Loss: 5.4782 | Val Loss: 4.4416
Val Accuracy -> Style: 67.64% | Artist: 56.92% | Genre: 68.39%
*** Saved new best model to /content/drive/MyDrive/ArtGAN_Results/checkpoints/best_model.pth ***

```

```

Epoch [2/30]
Learning Rates -> Backbone: 9.78e-06 | Head: 9.76e-04
Training: 66% 234/353 [02:56<01:44, 1.14it/s]Error loading ./wikiart/Post_Impressionism/vincent-van-gogh_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file
Training: 100% 353/353 [04:21<00:00, 1.35it/s]
Validating: 100% 148/148 [01:20<00:00, 1.85it/s]
Train Loss: 4.4615 | Val Loss: 4.0015
Val Accuracy -> Style: 73.02% | Artist: 65.84% | Genre: 70.98%
*** Saved new best model to /content/drive/MyDrive/ArtGAN_Results/checkpoints/best_model.pth ***

```

```

Epoch [3/30]
Learning Rates -> Backbone: 9.14e-06 | Head: 9.05e-04
Training: 90% 318/353 [03:57<00:19, 1.82it/s]Error loading ./wikiart/Post_Impressionism/vincent-van-gogh_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file
Training: 100% 353/353 [04:22<00:00, 1.34it/s]
Validating: 100% 148/148 [01:22<00:00, 1.80it/s]
Train Loss: 4.1409 | Val Loss: 3.8423
Val Accuracy -> Style: 74.70% | Artist: 69.77% | Genre: 71.21%
*** Saved new best model to /content/drive/MyDrive/ArtGAN_Results/checkpoints/best_model.pth ***

```

```
Epoch [4/30]
```

Learning Rates -> Backbone: 8.15e-06 | Head: 7.94e-04  
Training: 64% 227/353 [02:47<01:09, 1.80it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
Training: 100% 353/353 [04:19<00:00, 1.36it/s]  
Validating: 100% 148/148 [01:22<00:00, 1.79it/s]  
Train Loss: 3.9158 | Val Loss: 3.6808  
Val Accuracy -> Style: 75.82% | Artist: 72.53% | Genre: 73.68%  
\*\*\* Saved new best model to /content/drive/MyDrive/ArtGAN\_Results/checkpoints/best\_model.pth \*\*\*

Epoch [5/30]  
Learning Rates -> Backbone: 6.89e-06 | Head: 6.55e-04  
Training: 18% 64/353 [00:54<05:27, 1.13s/it]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
Training: 100% 353/353 [04:21<00:00, 1.35it/s]  
Validating: 100% 148/148 [01:20<00:00, 1.84it/s]  
Train Loss: 3.7125 | Val Loss: 3.5684  
Val Accuracy -> Style: 78.71% | Artist: 74.82% | Genre: 75.06%  
\*\*\* Saved new best model to /content/drive/MyDrive/ArtGAN\_Results/checkpoints/best\_model.pth \*\*\*

Epoch [6/30]  
Learning Rates -> Backbone: 5.50e-06 | Head: 5.01e-04  
Training: 88% 312/353 [03:51<00:31, 1.31it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
Training: 100% 353/353 [04:21<00:00, 1.35it/s]  
Validating: 100% 148/148 [01:17<00:00, 1.90it/s]  
Train Loss: 3.5475 | Val Loss: 3.5135  
Val Accuracy -> Style: 78.90% | Artist: 75.74% | Genre: 75.53%  
\*\*\* Saved new best model to /content/drive/MyDrive/ArtGAN\_Results/checkpoints/best\_model.pth \*\*\*

Epoch [7/30]  
Learning Rates -> Backbone: 4.11e-06 | Head: 3.46e-04  
Training: 7% 23/353 [00:22<05:43, 1.04s/it]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
Training: 100% 353/353 [04:21<00:00, 1.35it/s]  
Validating: 100% 148/148 [01:17<00:00, 1.91it/s]  
Train Loss: 3.3954 | Val Loss: 3.4657  
Val Accuracy -> Style: 79.43% | Artist: 76.46% | Genre: 76.04%  
\*\*\* Saved new best model to /content/drive/MyDrive/ArtGAN\_Results/checkpoints/best\_model.pth \*\*\*

Epoch [8/30]  
Learning Rates -> Backbone: 2.85e-06 | Head: 2.07e-04  
Training: 75% 265/353 [03:16<00:59, 1.48it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
Training: 100% 353/353 [04:19<00:00, 1.36it/s]  
Validating: 100% 148/148 [01:18<00:00, 1.88it/s]  
Train Loss: 3.2798 | Val Loss: 3.4069  
Val Accuracy -> Style: 79.52% | Artist: 77.71% | Genre: 77.06%  
\*\*\* Saved new best model to /content/drive/MyDrive/ArtGAN\_Results/checkpoints/best\_model.pth \*\*\*

Epoch [9/30]

Learning Rates -> Backbone: 1.86e-06 | Head: 9.64e-05  
Training: 21% 74/353 [00:59<03:06, 1.50it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
Training: 100% 353/353 [04:15<00:00, 1.38it/s]  
Validating: 100% 148/148 [01:16<00:00, 1.94it/s]  
Train Loss: 3.2160 | Val Loss: 3.3733  
Val Accuracy -> Style: 80.18% | Artist: 78.10% | Genre: 76.99%  
\*\*\* Saved new best model to /content/drive/MyDrive/ArtGAN\_Results/checkpoints/best\_model.pth \*\*\*

Epoch [10/30]  
Learning Rates -> Backbone: 1.22e-06 | Head: 2.54e-05  
Training: 58% 203/353 [02:28<01:23, 1.81it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
Training: 100% 353/353 [04:14<00:00, 1.39it/s]  
Validating: 100% 148/148 [01:14<00:00, 1.98it/s]  
Train Loss: 3.1756 | Val Loss: 3.3454  
Val Accuracy -> Style: 80.54% | Artist: 78.65% | Genre: 77.42%  
\*\*\* Saved new best model to /content/drive/MyDrive/ArtGAN\_Results/checkpoints/best\_model.pth \*\*\*

Epoch [11/30]  
Learning Rates -> Backbone: 1.00e-05 | Head: 1.00e-03  
Training: 85% 299/353 [03:37<00:44, 1.21it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
Training: 100% 353/353 [04:13<00:00, 1.39it/s]  
Validating: 100% 148/148 [01:15<00:00, 1.97it/s]  
Train Loss: 3.4802 | Val Loss: 3.5584  
Val Accuracy -> Style: 77.59% | Artist: 75.08% | Genre: 75.87%  
No improvement (1/7)

Epoch [12/30]  
Learning Rates -> Backbone: 9.94e-06 | Head: 9.94e-04  
Training: 29% 103/353 [01:13<02:26, 1.71it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
Training: 100% 353/353 [04:06<00:00, 1.43it/s]  
Validating: 100% 148/148 [01:14<00:00, 1.98it/s]  
Train Loss: 3.4796 | Val Loss: 3.5499  
Val Accuracy -> Style: 77.86% | Artist: 75.42% | Genre: 74.89%  
No improvement (2/7)

Epoch [13/30]  
Learning Rates -> Backbone: 9.78e-06 | Head: 9.76e-04  
Training: 91% 320/353 [03:45<00:19, 1.65it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
Training: 100% 353/353 [04:07<00:00, 1.43it/s]  
Validating: 100% 148/148 [01:14<00:00, 1.98it/s]  
Train Loss: 3.4063 | Val Loss: 3.5017  
Val Accuracy -> Style: 78.33% | Artist: 76.10% | Genre: 76.27%  
No improvement (3/7)

Epoch [14/30]  
Learning Rates -> Backbone: 9.51e-06 | Head: 9.46e-04  
Training: 14% 51/353 [00:38<03:47, 1.32it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken

data stream when reading image file  
 Training: 100% 353/353 [04:09<00:00, 1.41it/s]  
 Validating: 100% 148/148 [01:15<00:00, 1.96it/s]  
 Train Loss: 3.3365 | Val Loss: 3.4927  
 Val Accuracy -> Style: 78.97% | Artist: 76.21% | Genre: 76.04%  
 No improvement (4/7)

Epoch [15/30]  
 Learning Rates -> Backbone: 9.14e-06 | Head: 9.05e-04  
 Training: 66% 234/353 [02:45<01:27, 1.35it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
 Training: 100% 353/353 [04:09<00:00, 1.41it/s]  
 Validating: 100% 148/148 [01:15<00:00, 1.96it/s]  
 Train Loss: 3.2782 | Val Loss: 3.4495  
 Val Accuracy -> Style: 79.71% | Artist: 76.99% | Genre: 76.61%  
 No improvement (5/7)

Epoch [16/30]  
 Learning Rates -> Backbone: 8.68e-06 | Head: 8.54e-04  
 Training: 31% 108/353 [01:18<02:24, 1.69it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
 Training: 100% 353/353 [04:07<00:00, 1.43it/s]  
 Validating: 100% 148/148 [01:15<00:00, 1.95it/s]  
 Train Loss: 3.2179 | Val Loss: 3.4106  
 Val Accuracy -> Style: 79.33% | Artist: 78.52% | Genre: 76.55%  
 No improvement (6/7)

Epoch [17/30]  
 Learning Rates -> Backbone: 8.15e-06 | Head: 7.94e-04  
 Training: 14% 48/353 [00:36<03:53, 1.31it/s]Error loading ./wikiart/Post\_Impressionism/vincent-van-gogh\_l-arlesienne-portrait-of-madame-ginoux-1890.jpg: broken data stream when reading image file  
 Training: 100% 353/353 [04:11<00:00, 1.41it/s]  
 Validating: 100% 148/148 [01:15<00:00, 1.96it/s]  
 Train Loss: 3.1384 | Val Loss: 3.3935  
 Val Accuracy -> Style: 80.39% | Artist: 78.25% | Genre: 76.48%  
 No improvement (7/7)

Early stopping after 17 epochs.

=== Training Complete ===

Best Val Loss: 3.3454

Model saved at: /content/drive/MyDrive/ArtGAN\_Results/checkpoints/best\_model.pth

In [ ]: !pip install seaborn scikit-learn matplotlib



Requirement already satisfied: seaborn in /usr/local/lib/python3.12/dist-packages (0.13.2)  
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)  
Requirement already satisfied: matplotlib in /usr/local/lib/python3.12/dist-packages (3.10.0)  
Requirement already satisfied: numpy!=1.24.0,>=1.20 in /usr/local/lib/python3.12/dist-packages (from seaborn) (2.0.2)  
Requirement already satisfied: pandas>=1.2 in /usr/local/lib/python3.12/dist-packages (from seaborn) (2.2.2)  
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)  
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)  
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)  
Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.3.3)  
Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (0.12.1)  
Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (4.62.1)  
Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (1.5.0)  
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (26.0)  
Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (11.3.0)  
Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (3.3.2)  
Requirement already satisfied: python-dateutil>=2.7 in /usr/local/lib/python3.12/dist-packages (from matplotlib) (2.9.0.post0)  
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2025.2)  
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.2->seaborn) (2025.3)  
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)

```
In [ ]: %cd /content
        !python evaluate.py
```

```
/content
Using device: cuda
Results will be saved to: /content/drive/MyDrive/ArtGAN_Results/evaluation

Loading validation data...
Loaded 4707 val images with all 3 labels.
Loading trained model...
Loading checkpoint: /content/drive/MyDrive/ArtGAN_Results/checkpoints/best_model.
pth
Checkpoint from epoch 9, val_loss=3.345390876402726

=====
RUNNING COMPREHENSIVE EVALUATION
=====
Evaluating: 100% 148/148 [01:15<00:00, 1.95it/s]

=====
STYLE METRICS
=====
Top-1 Accuracy : 80.54%
Top-3 Accuracy : 94.69%
Top-5 Accuracy : 97.94%
Macro F1       : 36.85%
Macro Precision: 37.27%
Macro Recall   : 36.64%
Report saved   : /content/drive/MyDrive/ArtGAN_Results/evaluation/style_classif
ication_report.txt

=====
ARTIST METRICS
=====
Top-1 Accuracy : 78.65%
Top-3 Accuracy : 91.91%
Top-5 Accuracy : 95.30%
Macro F1       : 76.50%
Macro Precision: 77.56%
Macro Recall   : 76.38%
Report saved   : /content/drive/MyDrive/ArtGAN_Results/evaluation/artist_classi
fication_report.txt

=====
GENRE METRICS
=====
Top-1 Accuracy : 77.42%
Top-3 Accuracy : 96.60%
Top-5 Accuracy : 99.07%
Macro F1       : 66.82%
Macro Precision: 66.78%
Macro Recall   : 67.03%
Report saved   : /content/drive/MyDrive/ArtGAN_Results/evaluation/genre_classif
ication_report.txt

Metrics summary saved to /content/drive/MyDrive/ArtGAN_Results/evaluation/metrics
_summary.json

Generating confusion matrices...
Confusion matrix saved: /content/drive/MyDrive/ArtGAN_Results/evaluation/style_co
nfusion_matrix.png
Confusion matrix saved: /content/drive/MyDrive/ArtGAN_Results/evaluation/artist_c
onfusion_matrix.png
```

Confusion matrix saved: /content/drive/MyDrive/ArtGAN\_Results/evaluation/genre\_confusion\_matrix.png

Running outlier detection...

Found 50 high-confidence outliers.

Outlier JSON saved: /content/drive/MyDrive/ArtGAN\_Results/evaluation/outliers.json

Outlier report saved: /content/drive/MyDrive/ArtGAN\_Results/evaluation/outlier\_report.txt

=====

EVALUATION COMPLETE

=====

|       |  |            |  |              |  |              |  |           |
|-------|--|------------|--|--------------|--|--------------|--|-----------|
| STYLE |  | Acc: 80.5% |  | Top-3: 94.7% |  | Top-5: 97.9% |  | F1: 36.8% |
|-------|--|------------|--|--------------|--|--------------|--|-----------|

|        |  |            |  |              |  |              |  |           |
|--------|--|------------|--|--------------|--|--------------|--|-----------|
| ARTIST |  | Acc: 78.6% |  | Top-3: 91.9% |  | Top-5: 95.3% |  | F1: 76.5% |
|--------|--|------------|--|--------------|--|--------------|--|-----------|

|       |  |            |  |              |  |              |  |           |
|-------|--|------------|--|--------------|--|--------------|--|-----------|
| GENRE |  | Acc: 77.4% |  | Top-3: 96.6% |  | Top-5: 99.1% |  | F1: 66.8% |
|-------|--|------------|--|--------------|--|--------------|--|-----------|

Outliers found: 50

All results saved to: /content/drive/MyDrive/ArtGAN\_Results/evaluation/

In [ ]: